AMINE OUICHOU

📍 **Saint-Ouen, France | 📞 +33 6 07 76 21 23 | 📧 a.ouichou@gmail.com**

[LinkedIn] [GitHub] [Portfolio(WIP)]

PROFESSIONAL SUMMARY

Systems, Cloud & AI Software Engineer with production experience building HIPAA-compliant healthcare platforms and AI-native applications. Expert in architecting secure distributed systems using AI provider APIs (Chat, Embeddings, Function Calling, Agents) and modern cloud stacks (Zero Trust, mTLS). Combines deep low-level expertise (C/C++) with high-level AI orchestration, RAG architectures, and production-grade DevOps pipelines, from kernel-level to cloud-native.

EDUCATION

42 Paris:

Expert in IT Architecture Master's degree | 2022 - present

TECHNICAL SKILLS

- **Systems Programming:** C, C++, Linux Kernel Programming, Process/Thread Management, Signal Handling, IPC
- **AI & LLM Integration:** Mistral AI SDK (Chat, Embeddings, Agents, Function Calling), OpenAI-compatible APIs, RAG (pgvector), sentence-transformers, Prompt Engineering
- **Backend Development:** Python, FastAPI, Django, PostgreSQL, Redis, Domain-Driven Design, REST/WebSocket APIs
- **Cloud & DevOps:** Docker, Istio, GitHub Actions CI/CD, Multi-cloud Architecture (AWS, Azure, Cloudflare(CDN + R2), DigitalOcean, Heroku, Vercel)
- **Security Engineering:** Zero Trust Architecture, mTLS, Keycloak/OIDC, JWT/RBAC, Audit Logging, HIPAA Compliance, CSRF Protection.
- **Observability:** OpenTelemetry, Prometheus/Grafana, Structured Logging with Correlation IDs

PROFESSIONAL EXPERIENCE

Software Engineer Intern | Qynapse | [05/2025] - [11/2025]

- Architected and implemented a HIPAA-compliant clinical analytics platform using Zero Trust Architecture with Istio mTLS and Keycloak RBAC
- Developed a production-ready internal Keycloak integration library (2,000+ LOC) deployed across 2 applications, featuring async JWT validation with Redis caching
- Designed and built a Domain-Driven Design patient monitoring module with NIfTI medical imaging processing for multi-encounter brain scan comparisons
- Implemented comprehensive observability: OpenTelemetry tracing, structured audit logging with correlation IDs, and Prometheus metrics
- Created 30+ REST endpoints with 85% test coverage (55 integration tests) and full OpenAPI documentation
- Developed data visualization modules (radar, linear trend, box plot charts) using Next.js and ECharts for clinical cohort analysis

SELECTED PROJECTS

Mistral Realms – AI-Powered D&D Platform | Personal Project

- Built an 81,000-line full-stack platform using 4 Mistral API surfaces: Chat Completions, Function Calling (16 tools), Embeddings (RAG memory with pgvector), and Agents API (image generation)
- Developed a multi-provider AI architecture (6 providers) with automatic failover, context transfer, and agentic response validation against D&D 5e rules
- Backend: FastAPI, Python 3.13, SQLAlchemy 2.0, 42 services, 80+ endpoints, 10-layer middleware stack
- Frontend: Next.js, React 19, TypeScript, Tailwind CSS, shadcn/ui, i18n (EN/FR), lazy-loaded panels with streaming
- 1,930 tests (unit + integration + E2E Playwright), full CI/CD (6 GitHub Actions workflows), Docker multi-stage builds
- Production observability: OpenTelemetry tracing, 40+ Prometheus metrics, Grafana dashboards, structured logging
    - Live: https://realms.anguelz.tech
    - Code: https://github.com/aouichou/Realms

Personal Project | Cloud-Native Portfolio

- Full-stack deployment with Next.js and Django.
- Implemented full CI/CD with GitHub Actions, Codacy security scanning, and Dependabot automation
- Built a production-grade, multi-service architecture with Django Channels (WebSockets), Redis, and isolated Docker containers
- Deployed across multiple cloud providers (Heroku, DigitalOcean, Render) with Cloudflare CDN and R2 storage

  ➥ **Real-Time Cloud Terminal Service**

  - Implemented a secure proxy architecture: Browser → Django → FastAPI Terminal Service → PTY processes
  - Added comprehensive security: command allowlists, rate limiting (10 req/min), read-only filesystems, WAF protection

miniRT - Raytracing Engine | 42 Paris (Score: 125%)

- Implemented a ray tracing engine from scratch in C using MiniLibX, rendering 3D scenes with lighting and shadows
- Developed mathematical models for camera projections, light interactions, and geometric transformations
- Optimized rendering performance for complex scenes with multiple objects and light sources

ft_transcendence - Real-Time Web App | 42 Paris (Score: 125%)

- Developed a production-ready Pong tournament platform with Django backend and WebSocket gameplay
- Implemented OAuth2 authentication, 2FA, real-time matchmaking, and tournament brackets
- Containerized with Docker Compose and Nginx reverse proxy across multiple microservices